Simulation Arena

Deal Approval & Margin Balancing

Approve anything between 18% and 20% - 18% is the best single number. The 25% the rep is asking for is past it - that costs about $37k of expected profit.

Seed 42 | Run run_f2a692a6fa7e | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 21 Sep 2026, 20:10

Same seed, same numbers - this run can be reproduced exactly.

How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

The numbers that matter

BEST DISCOUNT TO OFFER
18%
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
$176k
Average profit if you made this call many times over.

DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY
18% to 20%
Across this range the expected profit is the same within the model's own margin of error. The model's own margin of error here is about $4k, so anywhere in this range is a judgement call, not a calculation.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
22%
At 22% the expected profit falls $16k below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $16k

EXPECTED PROFIT IF YOU CAVE IN
$139k
What 25% actually earns you.
vs -$37k vs the optimum

PROFIT GIVEN AWAY
$37k
Buying +15 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
77.6%
The chance the deal lands.

WORST CASE (1-IN-20)
$0
In an unlucky run of 20 deals, this is roughly the profit you would see.

How we got there

1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $10k straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $600k on $1.00M of list value

2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 15% discount is where they are most persuadable.

Evidence: Curve source: your assumption

3. The sweet spot

18% discount gives the highest expected profit at $176k, with a 77.6% chance of winning.

Evidence: peak of 17 levels tested

4. The range where the answer is the same

Across this range the expected profit is the same within the model's own margin of error. 18% is the best single number, but anything from 18% to 20% is a defensible call - that is where the rep has room to negotiate.

Evidence: margin of error $4k; safe range 18% to 20%

5. What the requested discount really costs

Moving from 20% to 25% buys +15 points of win probability but costs $37k of expected profit - more than the model's margin of error, so the loss is real.

Evidence: $139k vs $176k

Charts

Where profit peaks

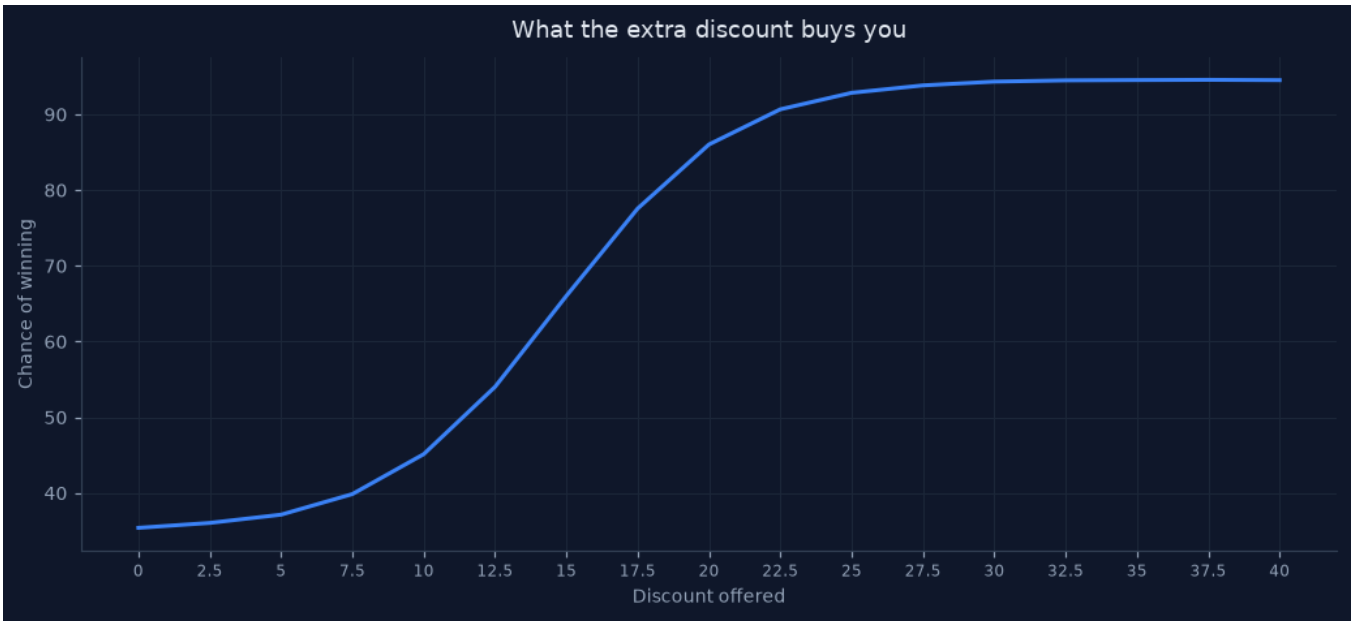
Everything from 18% to 20% gives the same answer within the model's margin of error; the peak is 18%



What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 18% to 20% range. Past that you are genuinely paying for the win.

What the extra discount buys you

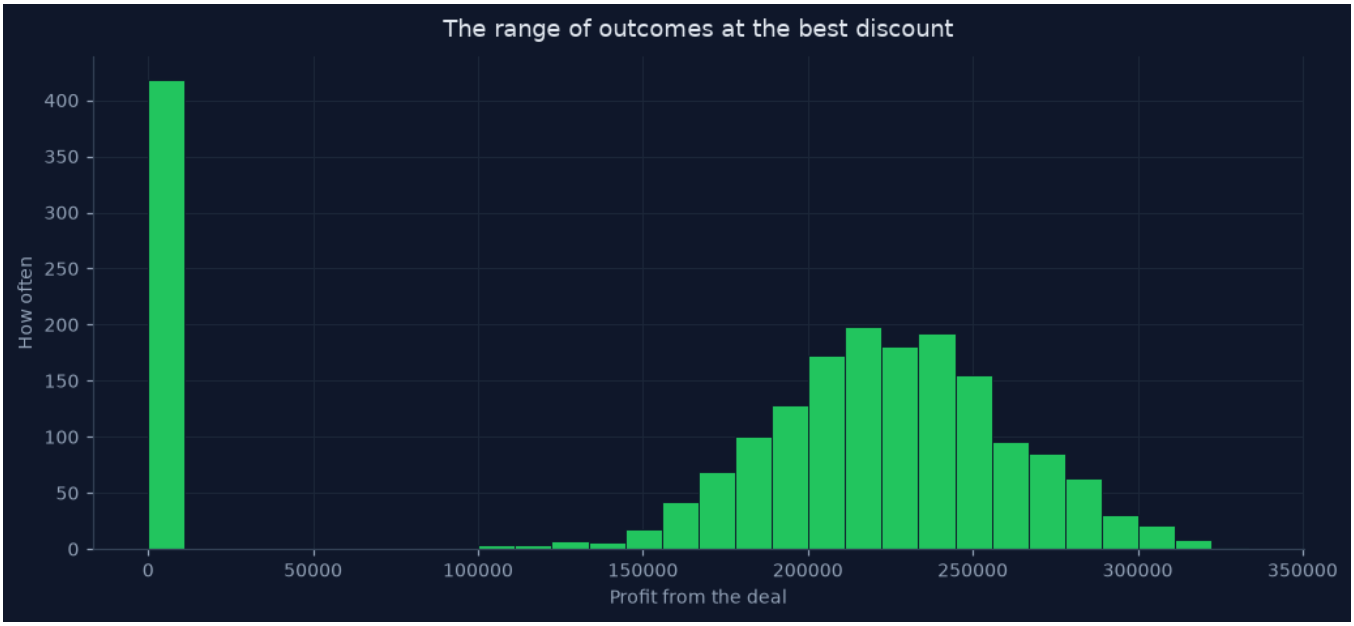
Curve learned from your assumption



What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.

The range of outcomes at the best discount

10,000 simulated worlds at 18% discount



What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.

Tables

Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
0.0%	$1,000,000	$399,934	35.4%	$143,251	$0	0% chance
2.5%	$975,000	$375,335	36.1%	$134,110	$0	0% chance
5.0%	$950,000	$349,972	37.1%	$130,921	$0	0% chance
7.5%	$925,000	$324,777	39.9%	$130,507	$0	0% chance
10.0%	$900,000	$299,848	45.1%	$134,435	$0	0% chance
12.5%	$875,000	$276,065	54.0%	$148,515	$0	0% chance

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
15.0%	$850,000	$250,196	66.0%	$165,362	$0	0% chance
17.5%	$825,000	$225,891	77.6%	$175,899	$0	0% chance
20.0%	$800,000	$200,428	86.0%	$173,135	$0	0% chance
22.5%	$775,000	$175,141	90.7%	$159,431	$0	1% chance
25.0%	$750,000	$149,937	92.9%	$138,513	$0	5% chance
27.5%	$725,000	$124,700	93.8%	$117,123	$0	14% chance
30.0%	$700,000	$99,154	94.3%	$93,522	$0	34% chance
32.5%	$675,000	$74,814	94.5%	$70,346	$0	57% chance
35.0%	$650,000	$49,418	94.5%	$46,929	-$8,278	79% chance
37.5%	$625,000	$25,567	94.6%	$24,234	-$31,716	92% chance
40.0%	$600,000	-$22	94.5%	-$66	-$57,942	98% chance

Row 8 maximises expected profit.

The decision in one line

What changes if you approve the discount being asked for.

Column 1	Best (18%)	Asked (25%)	Difference
Discount	17.5%	25.0%	+7.5 pts
Win chance	77.6%	92.9%	+15.3 pts
Profit if won	$225,891	$149,937	-$75,954
Expected profit	$175,899	$138,513	-$37,386
Range where the answer is the same	18% to 20%	outside it	margin of error $4k

The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

Discount	Expected profit	Short of the best	Inside the range?	What it means
0.0%	$143,251	$32,649	Outside	costs you $33k
2.5%	$134,110	$41,789	Outside	costs you $42k
5.0%	$130,921	$44,978	Outside	costs you $45k
7.5%	$130,507	$45,392	Outside	costs you $45k
10.0%	$134,435	$41,464	Outside	costs you $41k
12.5%	$148,515	$27,384	Outside	costs you $27k
15.0%	$165,362	$10,537	Outside	costs you $11k
17.5%	$175,899	$0	Inside	safe to approve
20.0%	$173,135	$2,764	Inside	safe to approve
22.5%	$159,431	$16,468	Outside	costs you $16k
25.0%	$138,513	$37,386	Outside	costs you $37k
27.5%	$117,123	$58,776	Outside	costs you $59k
30.0%	$93,522	$82,377	Outside	costs you $82k
32.5%	$70,346	$105,553	Outside	costs you $106k
35.0%	$46,929	$128,970	Outside	costs you $129k
37.5%	$24,234	$151,665	Outside	costs you $152k
40.0%	-$66	$175,965	Outside	costs you $176k

Highlighted rows are the range where the answer is the same (18% to 20%) - within about $4k of the best.

What we recommend

Approve anything between 18% and 20% - 18% is the best single number. The 25% the rep is asking for is past it - that costs about $37k of expected profit.

Risk Low 25 out of 100 | Confidence: Medium

Why

• Any discount from 18% to 20% gives the same expected profit within the model's own margin of error, so there is real room to negotiate there.

• Expected profit peaks at 18% ($176k).

• The requested 25% delivers $139k - $37k less than the peak, which is more than the model's margin of error.

• Win probability only improves +15 points for that giveaway.

What to do

1. Approve any discount between 18% and 20% - they are the same answer within the model's own margin of error.

2. Above 20% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $7k per point of discount.

3. If this quarter must not misfire, 0% is the safer call: it maximises the bad case rather than the average.

The trade-off: You are trading $37k of expected profit for +15 points of win probability.

What we assumed and what to watch

Worth knowing before you act

• No historical deal file was mapped, so the buyer-response curve is your assumption. Upload past deals to learn it from evidence.

Assumptions

• List value $1,000,000, cost $600,000 (40.0% standard margin).

• Consumption effects: none - your file has no tier accelerator, burst or rollover columns, so this is a straight price-versus-margin question.

• Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 14.8% discount (source: your assumption).

• Win probability uncertainty: plus or minus 5 points.

• Cost volatility: plus or minus 6% of cost.

• Margin floor: 12.0%.

• Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $4k at the best one), or smaller than 1% of the best profit - whichever is larger.

• The range where the answer is the same is 18% to 20%, taken from the 17 discount levels we tested.

• 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).

• Deal / quote ID -> not in your file; not used

• Customer name -> not in your file; not used

• Deal value (list price) -> not in your file; used the standard value (1000000) instead

• Cost to deliver -> not in your file; not used

• Standard margin (%) -> not in your file; used the standard value (0.4) instead

• Discount given (%) -> not in your file; not used

- Won? (1/0) -> not in your file; not used
- Hit a consumption tier accelerator -> not in your file; not used
- Unmetered burst used -> not in your file; not used
- Rollover window -> not in your file; not used

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Discount given (%) -> not in your file; not used
0.00s	prepare	Won? (1/0) -> not in your file; not used
0.00s	prepare	Hit a consumption tier accelerator -> not in your file; not used
0.00s	prepare	Unmetered burst used -> not in your file; not used
0.00s	prepare	Rollover window -> not in your file; not used
0.00s	prepare	Checking your assumptions…
0.00s	simulate	Running the simulation…
0.00s	read	Reading the deal you gave us...
0.00s	read	Working from the numbers you typed: $1.00M at 40.0% margin.
0.00s	prepare	Cost to deliver is about $600k.
0.00s	prepare	Working out how much a discount really moves the buyer...
0.00s	prepare	No historical win/loss columns mapped - using the curve you set.
0.00s	simulate	Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)...
0.00s	simulate	Simulated 11% of the discount levels...
0.00s	simulate	Simulated 23% of the discount levels...
0.00s	simulate	Simulated 35% of the discount levels...
0.01s	simulate	Simulated 47% of the discount levels...
0.01s	simulate	Simulated 58% of the discount levels...
0.01s	simulate	Simulated 70% of the discount levels...
0.01s	simulate	Simulated 82% of the discount levels...
0.01s	simulate	Simulated 94% of the discount levels...
0.01s	simulate	Simulated 100% of the discount levels...
0.01s	summarize	All discount levels tested. Looking for the sweet spot...
0.01s	summarize	Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 18% to 20%.
0.01s	summarize	Done. Best discount is 18% ($176k expected profit); the range where the answer is the same is 18% to 20%.